

ESP32-S3-N16R8 — DIRECT 120VAC WIRING TO DOMETIC BRISK II

SSR-40DA for Compressor | Relay Channels for Fans + Furnace | Dual Freeze Protection

ESP32-S3-N16R8 + 4-CH RELAY

44-Pin Dev Board | 5V USB-C
Relay: 10A, Active HIGH (jumper)

CH1 (GPIO 4): Furnace

CH2 (GPIO 5): Compressor→SSR

CH3 (GPIO 6): Fan Low

CH4 (GPIO 15): Fan High

Reset Btn: GPIO 2

RGB LED: GPIO 48

I2C: GPIO 41/40

DS18B20: GPIO 42

GND

12V+

3.3V

FURNACE

(Dry Contact Closure)

Separate unit with own blower

T1

T2

SSR-40DA

Solid State Relay
3-32VDC → 25A/380VAC

DC+

DC-

CH1 NO

120VAC HOT

HOT

NEUTRAL

AC1 (in)

AC2 (out)

LOAD CHAIN

Supco SFPC (freeze stat)

NC: opens 35F, closes 50F

Bimetal Cutout

Fan Hi (CH4 → Pin 2)

Fan Lo (CH3 → Pin 4)

120VAC SOURCE

6-PIN DOMETIC CABLE

to Brisk II Rooftop AC
(14 AWG minimum)

Pin 1: Blue (Compressor)

Pin 2: Black (Fan Hi)

Pin 3: Yellow (unused)

Pin 4: Red (Fan Lo)

Pin 5: White (Neutral)

Pin 6: Green (Ground)

DS18B20 FREEZE SENSOR

VCC Waterproof probe on evaporator coil

Data Software: Cut compressor @ 32F, restart @ 45F

GND

4.7KΩ pull-up: Data → 3.3V

SAFETY CRITICAL — 120VAC WIRING

CH2 relay NO switches 12VDC milliamp trigger to SSR DC+
SSR-40DA switches 120VAC to compressor (heatsink required ~15W @ 12A)
CH3/CH4 switch 120VAC directly to fans (2-3A, 10A relay rating)
CH1 = dry contact for furnace (separate unit)
Dual freeze protection: DS18B20 software (32F) + Supco SFPC hardware (35F)
All 120VAC wiring: 14 AWG minimum, proper junction box per NEC

COMPONENT SPECIFICATIONS

ESP32-S3-N16R8 + 4-CHANNEL RELAY MODULE:

ESP32: 16MB Flash, 8MB PSRAM | 5V USB-C
Relay: 10A per channel, Active HIGH (4 channels)

GPIO ASSIGNMENTS:

CH1=GPIO4 CH2=GPIO5 CH3=GPIO6 CH4=GPIO15
I2C: SDA=GPIO41, SCL=GPIO40 | 1-Wire: GPIO42
RGB LED: GPIO48 | Reset Button: GPIO2 (NO to GND)